VEDANT ANANT KULKARNI

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AUTONOMOUS ROBOTICS | ROS 2 | SLAM | NAVIGATION

PROFESSIONAL SUMMARY

Electronics and Telecommunication Engineering undergraduate specializing in autonomous robotics. Builds ROS 2-based systems spanning SLAM, Nav2 navigation, LiDAR and depth-camera perception, manipulator trajectory planning, PX4 UAV flight control, and embedded hardware integration. Former Product Developer Intern at Robobox.

EDUCATION

B.E., Electronics and Telecommunication Engineering **2024 – Present**
Bharati Vidyapeeth's College of Engineering, Navi Mumbai (University of Mumbai)
Semesters I–IV: 68.37%, 69.29%, 68.26%, 70.62% | Top 20% of class
HSC, Maharashtra State Board – 71.83% **2024**
SSC, Maharashtra State Board – 98.20% (School Topper, Rank 1 of 550) **2022**

TECHNICAL SKILLS

- **Robotics Software Engineering:** Modular ROS 2 architecture (nodes, topics, services, actions), QoS tuning, TF2 frame management, URDF/Xacro robot modelling, ros2_control, parameterised launch files
- **Autonomy and Navigation:** Layered navigation design (global and local planning, costmaps), LiDAR SLAM and localization (Cartographer, Nav2), odometry and IMU sensor fusion, waypoint-based mission execution
- **Motion Planning and Flight Control:** Kinematics and trajectory planning, collision-aware manipulation (MoveIt 2), PX4 offboard control with state-estimation integration
- **Simulation and Validation:** Simulation-first development and incremental testing in Gazebo, MuJoCo, and PX4 SITL; sim-to-real transfer
- **Perception:** Camera calibration, image-processing pipelines, fiducial marker detection (ArUco), face detection and recognition, pose estimation (OpenCV)
- **Embedded Integration:** Hardware–software interfacing, motor control with encoder and IMU feedback (Raspberry Pi, ESP32, STM32, Arduino)
- **Engineering Practices:** Python, Bash, Linux, Git/GitHub version control, systematic debugging with RViz and ROS 2 CLI tools

PROJECTS

Strata Cobot – Collaborative Multi-Robot System
Robotic Arm, AMR, Depth Cameras, Custom Planning Algorithms
- Developed a multi-robot system in which a robotic arm and an autonomous mobile robot (AMR) platform work collaboratively on shared tasks.
- Computed arm trajectories with singularity avoidance for smooth and reliable manipulation.
- Added a perception layer using depth cameras; implemented AMR global and local path planning for navigation in the given environment.
- Wrote all planning algorithms from scratch, using no planning framework or stack, and optimised them for efficiency.

CareBro Assist – Autonomous Assistive Mobile Robot
ROS 2, Nav2, Raspberry Pi, ESP32
- Implemented autonomous indoor navigation with path planning, waypoint navigation, and obstacle avoidance using Nav2.
- Developed LiDAR-based SLAM and localization, with OpenCV face recognition and LiDAR obstacle perception.
- Integrated Raspberry Pi and ESP32 for motor control and ROS 2 communication across the full autonomy stack.

Low-Cost 6-DOF Robotic Manipulator
ROS 2, MoveIt 2, OpenCV
- Designed a modular arm from scratch, including joint architecture, actuator selection, and interchangeable end-effectors; prototype under development.
- Configured MoveIt 2 for kinematics, trajectory generation, and collision-aware planning, with camera-based vision-guided pick-and-place.

PROFESSIONAL EXPERIENCE

Robobox **Jan 2026 – Jul 2026**
Product Developer Intern
- Developed robotic product features on the ROS 2 stack, covering simulation, software development, and system integration.
- Built and validated robot behaviours in Gazebo using URDF/Xacro models, TF2, and RViz under a simulation-first workflow.
- Applied advanced robotics practices in navigation, SLAM, sensor integration, and hardware–software interfacing to prototype and refine product capabilities.

Competitions and Recognition
- **e-Yantra Robotics Competition (2024, 2025):** Two consecutive years of participation in robotics, embedded systems, and programming challenges.
- **BVHacks 2025:** 3rd Prize overall and 1st Position in the Hardware Category.
- **Hackathon Finalist:** Reached the finals of three hackathons with hardware-based solutions.
- **INSPIRE Award 2022** (Department of Science & Technology, Govt. of India): Personal Assistive Robot for Physically Challenged People.
- **1st Prize, Interschool Science Project Competition 2022** (IUCAA and SP Foundation): Advanced Robots for Industrial Automation.